



Validation of OpenFoam for heavy gas dispersion applications



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HIGHLIGHTS

- Modification of OpenFoam to handle heavy gas dispersion.
- Validation of OpenFoam for heavy gas dispersion in low and high turbulence environment.
- Code to code comparison with Fluent.
- Gravity driven flow and atmospheric boundary layer flow including terrain effects were taken as reference cases.

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ABSTRACT

In the present paper heavy gas dispersion calculations were performed with OpenFoam. For a wind tunnel test case, numerical data was validated with experiments. For a full scale numerical experiment, a code to code comparison was performed with numerical results obtained from Fluent. The validation was performed in a gravity driven environment (slope), where the heavy gas induced the turbulence. For the code to code comparison, a hypothetical heavy gas release into a strongly turbulent atmospheric boundary layer including terrain effects was selected. The investigations were performed for SF₆ and CO₂ as heavy gases applying the standard $k-\epsilon$ turbulence model. A strong interaction of the heavy gas with the turbulence is present which results in a strong damping of the turbulence and therefore reduced heavy gas mixing. Especially this interaction, based on the buoyancy effects, was studied in order to ensure that the turbulence-buoyancy coupling is the main driver for the reduced mixing and not the global behaviour of the turbulence modelling.

For both test cases, comparisons were performed between OpenFoam and Fluent solutions which were mainly in good agreement with each other. Beside steady state solutions, the time accuracy was investigated. In the low turbulence environment (wind tunnel test) which for both codes (laminar solutions) was in good agreement, also with the experimental data. The turbulent solutions of OpenFoam were in much better agreement with the experimental results than the Fluent solutions. Within the strong turbulence environment, both codes showed an excellent comparability.

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1. Introduction

The accidental release of liquid or gaseous toxic or flammable hazardous materials can form a large potential risk to the environment and people. In risk analysis, for certain situations or installations a large amount of failure scenarios, release and weather conditions are assessed in order to estimate the potential risk to the public. Materials like ammonia, LPG, LNG but also CO₂ are transported via road or train in large amounts. Accidentally released, they can be dispersed showing strong heavy gas behaviour. Usually, gas is considered as a heavy gas when its relative density is larger than 1.15 with respect to air at ambient temperature [1]. This leads to gravitationally driven flows which

can result in large upwind and lateral dispersion and large density gradients in vertical direction. This negative density gradient leads to a locally strongly stratified boundary layer which dampens the turbulence significantly. Therefore, the dispersion behaviour of heavy gases is significantly different compared with neutrally buoyant gases. In addition, released materials can undergo a transition in the dispersion behaviour from heavy gas to even negatively buoyant like, e.g. ammonia. Released in two phase as liquid droplets and gas, the droplets have to be vaporized. The required heat transfer, originating from the surrounding air, decreases the temperature of the air with the consequence of increasing the air density. This air then mixes with the hazardously releases material and might show heavy gas behaviour even at concentrations lower than 5%. During the dispersion process, the mixture is retaining heat again and is finally getting a mixture lighter than air. Since the initial heavy gas behaviour release is dominating the dispersion process, a correct modelling of the mixing and heat transfer is

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crucial in the modelling of hazardous materials. Therefore, heavy gas can occur with different scenarios, with gases heavier than air (like CO₂ [2]), 2-phase release of materials also lighter than air (e.g. ammonia [2–4], hydrogen [5]) or the release at low or even cryogenic temperature (e.g. LNG [6,7]) and is one of the main interests in dispersion modelling.

For CFD dispersion calculations, commercial software packages are widely used [2–7] since they are user friendly and commonly accepted within the community. For atmospheric boundary layer flow, these codes (e.g. Fluent, CFX) are considered as validated since extensive research was done especially at universities including RANS but also LES modelling [8,9]. For heavy gas applications, good agreement between the experimental results of, e.g. the Thorney Island experiment (Trial 8) and the numerical results obtained with Fluent are documented [10]. Usually, the commercial codes offer limited accessibility to incorporate own modules or modifications, therefore own code developments or, as an alternative, open source codes are used as alternatives. The open source software OpenFoam is meanwhile widely used especially for incompressible flow problems, also for atmospheric boundary layer flows [11–14], also including terrain effects [15] or buoyant atmospheric boundary layers [16]. So far, no publications were found on heavy gas applications. Nevertheless, it has to be ensured that the code is verified and validated for the targeted flow domain. The present paper is focusing on validation and code to code comparison particularly for heavy gas dispersion. In contrast to common atmospheric boundary layer flow, the interaction of the atmospheric boundary layer with the released heavy gas is strongly coupled via the turbulence model, buoyancy and gravitational effects. So far, OpenFoam seems not to be applied on heavy gas dispersion problems with the presence of multiple species at high mass fractions including gravitational effects. Within the EU project COCATE, TNO investigated CO₂ worst case pipeline rupture scenario's in realistic environments and performed dispersion calculations for the released heavy gas [10]. As part of this work, OpenFoam and Fluent were validated wrt. turbulence modelling and applied for heavy gas dispersion including terrain effects. The present paper will focus on the validation of heavy gas dispersion in a low turbulence environment and a code to code comparison between Fluent and OpenFoam in a high turbulence environment.

2. Numerical modelling

2.1. The open source OpenFoam software package

OpenFoam [17] is a fully open source software and it is meanwhile widely used for many applications usually in the incompressible, low Mach number flow domain. OpenFoam is delivered with numerous pre-configured solvers, utilities and libraries and so can be used like any typical simulation package. OpenFoam is open, both in terms of source code and in its structure and hierarchical design. This makes the solvers, utilities and libraries fully extensible. The software package was rather defined as toolbox than a universal code; this implies that the user also is responsible for the verification and validation of the tools being used. The classical CFD tools included in OpenFoam are all finite volume approaches and a large variety of turbulence models for RANS and LES are available. Most of the OpenFoam tools are designed as time accurate codes with global time stepping schemes which makes the codes usually quite slow which is widely reported in the community [11,13]. For steady state solutions, relaxation can be applied and the convergence is accelerated.

Beside a structured mesh generator which is included in the distribution and applicable for simple geometries, most of the common mesh formats can be read in or converters are available.

Therefore, code to code comparisons can be performed on identical meshes generated by other software packages quite easily.

2.2. ReactingFoam

The present investigation focuses on the ReactingFoam code (based on OpenFoam-2.1.0), which is the only one taking different species, chemical reactions, heat transfer and variable thermodynamic properties into account which is required in the targeted flow domain considered in external safety issues. Comparable with the features in Fluent, the solver is pressure based for low Mach number flow whereas the thermodynamic properties and also the turbulence model include compressibility effects. For the present calculations the equations for the chemical reactions were disabled. The standard $k-\epsilon$ turbulence model is chosen since this is the most commonly used one in wind engineering problems using a RANS approach. All convective and viscous terms are discretized 2nd order accurate using upwind or central schemes. For the temporal discretization the 1st order Euler scheme is chosen; since the time step restrictions due to the global time stepping schemes are forcing small time steps sufficient accuracy is ensured.

2.2.1. Thermodynamic and transport properties

Temperature dependant heat capacity, enthalpy and entropy are defined via temperature dependant polynomials for each species; the data format is based on the CHEMKIN format. The transport properties of the species are calculated via the Sutherland law and are by default predefined to standard values of air [18].

In order to be comparable with Fluent, where the molecular diffusion coefficient is specified explicitly and is not calculated via the mixture viscosity and a laminar Schmidt number, the species equation in ReactingFoam was modified such that the laminar diffusion terms are:

$$\vec{J}_i = -D_{i,m} \rho \nabla Y_i \quad (1)$$

$$\text{with } D_{i,m} \cdot \rho = \frac{\mu}{Sc_{lam}} \quad (2)$$

2.2.2. Gravity

The gravitational term included in the momentum equation is defined in a compressible way as:

$$\vec{F}_G = \rho \cdot \vec{g} \quad (3)$$

Since the targeted heavy gas dispersion takes place close to the ground, the aerostatic pressure in the atmospheric boundary layer is neglected and by this the vertical gradients in temperature and density are set to zero. This is done by applying the Boussinesq approximation, a reference density was introduced and the gravitational forces only play a role in regions with density gradients:

$$\vec{F}_G = (\rho - \rho_{ref}) \cdot \vec{g} \quad (4)$$

2.2.3. Turbulence modelling

Within the ReactingFoam solver, which is based on a compressible formulation and therefore the density is also effected by the pressure, the density is included in all flux-updates. For the turbulence model, there are incompressible and compressible versions available, for ReactingFoam the compressible ones are used which include also the density in all terms (standard $k-\epsilon$ model). For the present investigation in low speed flow, the compressible terms in the turbulence model are negligible since they are close to zero.

The equations for the turbulence kinetic energy and dissipation for the k - ε model reduce to:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \varepsilon \quad (5)$$

$$\begin{aligned} \frac{\partial}{\partial t}(\rho \varepsilon) + \frac{\partial}{\partial x_i}(\rho \varepsilon u_i) = & \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + C_{\varepsilon 1} \frac{\varepsilon}{k} (G_k + C_{\varepsilon 3} \cdot G_b) \\ & - C_{\varepsilon 2} \rho \frac{\varepsilon^2}{k} \end{aligned} \quad (6)$$

Here, G_k is the production term of turbulence kinetic energy due to the strain-rate tensor and G_b the buoyancy term:

$$G_b = -g_i \frac{\mu_t}{\rho Pr_t} \frac{\partial \rho}{\partial x_i} \quad (7)$$

The constants are given as $C_{\varepsilon 1} = 1.44$, $C_{\varepsilon 2} = 1.92$, $C_{\varepsilon 3} = -0.33$ and the turbulent Prandtl numbers for k and ε $\sigma_k = 1.0$ and $\sigma_\varepsilon = 1.3$. The turbulent Prandtl number Pr_t is 0.85.

Within Fluent [19], the G_b term in the ε equation is recommended to be switched off ("full buoyancy off", $C_{\varepsilon 3} = 0$). If the term is activated ("full buoyancy on"), not the original G_b term is applied but a modified one where the term is scaled by a factor which is maximized ($C_{\varepsilon 3} = 1$) when the local flow vector and the gravitational vector are parallel:

$$C_{\varepsilon 3} = \tanh \left| \frac{\vec{u} \cdot \vec{g}}{|\vec{u}| |\vec{g}|} \right| \quad (8)$$

Whether the form of $C_{\varepsilon 3}$ (Eq. (8)), which is targeting main flow directions or buoyant shear layers aligned with the gravity vector, is appropriate for heavy gas dispersion will not be discussed here. The treatment of the buoyancy-term in the ε -equation is still a point of discussion in heavy gas turbulence modelling research. In order to match specific flow features, most of the commercial codes (like Fluent and CFX) have individual treatment of the buoyancy term in the ε -equations [19,20].

The buoyancy term G_b is of major importance with heavy gas dispersion, therefore it has been incorporated in OpenFoam since it is not present in the distributed version. All three versions have been implemented in OpenFoam and compared: the buoyancy term modifying only the k -equation, the Fluent approach of modifying the ε equation (Eq. (8)) and the original k - ε formulation of applying the buoyancy term in the ε -equation with a constant $C_{\varepsilon 3}$ (Eq. (6)). The differences are discussed later in the paper.

Taking species mass diffusion into account, the turbulent viscosity is corrected with the turbulent Schmidt number. In OpenFoam, it is hard-wired with a value of 1.0 whereas typical values are between 0.5 and 1.4 depending on the application. Therefore, a freely selectable turbulent Schmidt number was implemented and the default value set to 0.7. A turbulent Prandtl number of 0.85 is applied for both Fluent and OpenFoam calculations including heat transfer.

3. Validation of ReactingFoam for a low turbulence, gravity driven dense gas release

3.1. Test case description DAT632 (University of Hamburg)

The tests were performed at the University of Hamburg, the flow for this experiment (DAT632) is purely driven by the gravitational effects of the released heavy gas (SF_6). The flow is quasi laminar due to the low flow velocities ($Re_l \sim 15,000$). Therefore, the numerical turbulence modelling can be tested wrt. the production of turbulence and the destruction due to the vertical density gradient in the

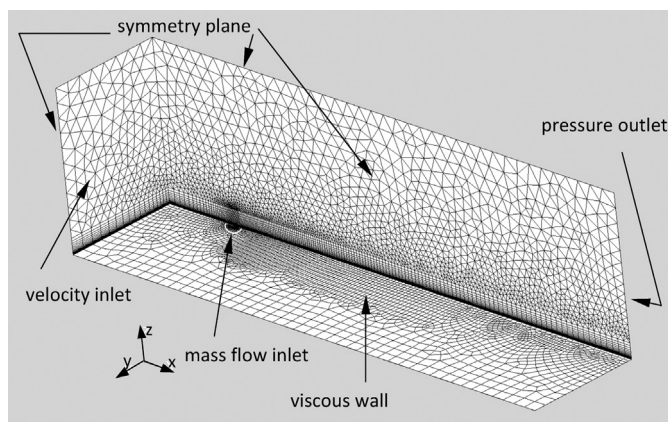


Fig. 1. Hybrid hexahedral-dominant computational grid and boundary conditions.

transition regime. The test case was also rebuilt numerically before with fair agreement with the experimental results [21]. The well documented experimental data is available within the REDIPHEN database [22].

The test section of the open-circuit wind tunnel at the University of Hamburg has the dimensions of $1.5 \text{ m} \times 1.0 \text{ m} \times 4 \text{ m}$ (width $\times$ height $\times$ length) [22]. The gas was released via a 70 mm plate containing numerous holes arranged concentrically in 4 rings such that a homogenous release flow is achieved. The release gas is SF_6 with a density ratio of 5.14 compared to air. The slope was oriented downward at 8.6%. The measurements were taken at discrete points on the floor with aspirated hot-film probes. Within the REDIPHEN database, all time dependant sensor data is available.

3.2. Numerical results for test case DAT632

The 3D meshes of the measurement section were generated as hexahedra-dominant hybrid meshes. The hexahedral layer consists of 32 layers, the overall number of cells is 160,000 for the standard mesh (Fig. 1). The maximum aspect ratio is 1400. Heavy gas dispersion simulations need a high wall normal resolution in order to achieve accurate numerical results. Therefore, high aspect ratio cells will be present on the largest surface cells. As long as the flow is aligned with the cells orientation, aspect ratios up to 20,000 are commonly used for heat transfer problems [23]. The maximum grid skewness is 2.3, the mesh non-orthogonality is 12.8 (averaged) with a maximum of 67. Nevertheless, the heavy gas is mainly dispersed in the hexahedral cells of the grid where non-orthogonality and skewness are lower. The computational domain is 1 m high, 0.78 m wide (applying symmetry conditions) and 3.5 m long, with the release point located at a distance of 0.5 m from the boundary. Instead of rotating the grid in order to achieve the required slope angles, the gravitational vector was adjusted such that the resulting vector is at the required angle. The heavy gas is released by a semi-circular boundary with 70 mm diameter as mass flow inlet. The mass flow is 0.08715 g/s (steady state for a long time in the range of minutes), corresponding to 0.1743 g/s for the full domain. In both lateral planes symmetry boundary conditions, at the upwind plane a velocity inflow boundary condition and at the downwind boundary an exit pressure boundary conditions are applied. At the upper boundary a slip boundary condition and at the surface a no-slip boundary condition is applied. The temperature is constant at all boundaries (300 K). For the turbulent calculations, turbulence kinetic energy is fixed at $10^{-6} \text{ m}^2/\text{s}^2$ and turbulence dissipation rate at $10^{-4} \text{ m}^2/\text{s}^3$. Since the wind tunnel is not operating during the test, the initial velocity for the experiment is zero in

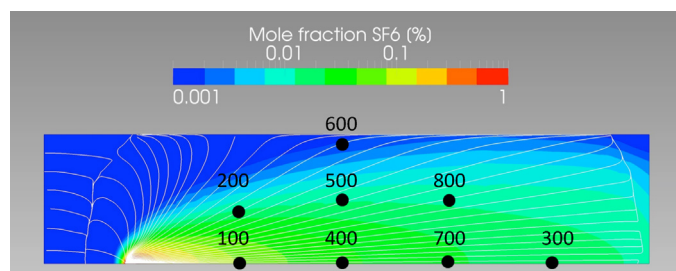


Fig. 2. Wall molar fraction of SF₆ including surface streamlines (Fluent solution) and sensor positions (black points, sensor numbers 100–800).

the whole domain. For stability reasons the initial flowfield velocity and the velocity at the inflow boundary condition is set to 10^{-3} m/s.

The SF₆ properties are given as:

$$\rho = 5.9 \text{ kg/m}^3 \text{ at } 300 \text{ K}$$

$$\mu = 1.72 \cdot 10^{-5} \text{ Ns/m}^2 \quad (\nu = 2.74 \cdot 10^{-6} \text{ m}^2/\text{s})$$

$$D = 3.9 \cdot 10^{-6} - 7.3 \cdot 10^{-6} \text{ m}^2/\text{s}$$

Due to the low gravitationally induced flow velocities and the damping of the turbulence due to the vertical density gradient inside the heavy gas the flow remains quasi laminar ($Re_l \sim 15,000$). Therefore the molecular mass diffusion plays a significant role. Calculations were performed with a variation in the laminar diffusion coefficient of 3.9 and $7.3 \cdot 10^{-6} \text{ m}^2/\text{s}$ as lower and upper limits from experimental data [24]. This corresponds to a laminar Schmidt number between 0.75 and 0.38 .

In the Fluent calculations, air was modelled as single species, whereas in ReactingFoam two species N₂ and O₂ were modelled (with mass fraction of 0.76 for N₂ and 0.24 for O₂). With ReactingFoam, the material properties were given via the CHEMKIN database.

The calculations both for Fluent and OpenFoam were performed with unsteady timestepping since the velocity (cloud arrival times at sensors) and the maximum concentrations give a good indication of the flow phenomena. In first instance, the steady state solutions are discussed.

A typical flow pattern of a laminar calculation (Fluent) of the 8.6° slope is shown in Fig. 2. Here, also the sensor locations and numbering is given. The released heavy gas is accelerated by gravitational forces down the slope. Due to the heavy gas behaviour the crosswind and also upwind spreading is high. This leads to rather high SF₆ concentrations also in lateral direction. In the main flow direction, the SF₆ is mixed rapidly. The maximum flow velocity inside the domain is 0.23 m/s.

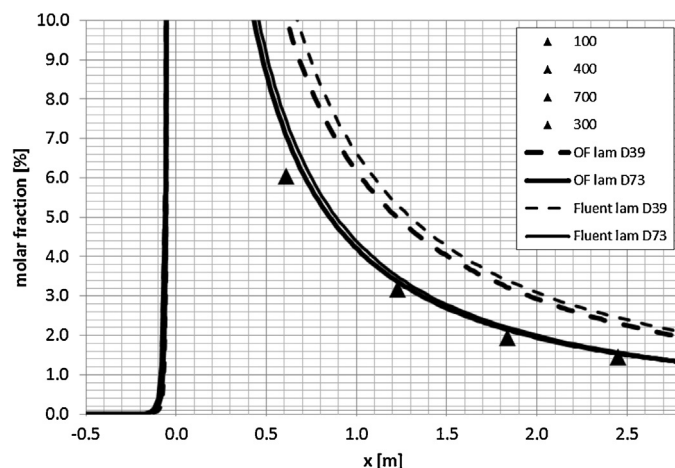


Fig. 3. Results at symmetry plane for laminar calculations of DAT632.

3.2.1. DAT 632: Laminar solutions

In a first step, only laminar flow was investigated in order to exclude possible sensitivities of the turbulence model.

The laminar solutions of Fluent and OpenFoam (Fig. 3) agree well for both laminar diffusion coefficients and the agreement with the experimental data is excellent for the higher diffusion coefficient ($D = 7.3 \cdot 10^{-6} \text{ m}^2/\text{s}$, $Sc = 0.38$). The solutions of OpenFoam show a minor offset of 0.1 – 0.2% ($D = 7.3 \cdot 10^{-6} \text{ m}^2/\text{s}$) and 0.2 – 0.4% ($D = 3.9 \cdot 10^{-6} \text{ m}^2/\text{s}$) to the reference solution of Fluent. Immediately behind the release, the molar fractions are slightly overestimated (at sensor 100 about 1%) but regarding the steep gradient the agreement is still considered good.

At the lateral sections $y = 0.31 \text{ m}$ and $y = 0.38 \text{ m}$ (Fig. 4), some minor differences occur between the two codes but the trends of the different solutions are the same. Keep in mind that there was an offset between the solutions of the different codes (0.1 – 0.4%) which is in the range of the differences discussed here. The agreement with the experimental data at sensor 200 (which is located at $y = 0.31 \text{ m}$) is excellent for the Fluent solution, whereas a minor difference occurs for the OpenFoam solution, which shows an increase in molar fractions about 0.1 m upstream. At the sensor locations 500 and 800 (located at $y = 0.38 \text{ m}$) the absolute value is slightly overestimated, but the gradient (both sensors show almost the same concentration) is predicted well for all solutions which show a wide local maximum at his location.

At the crosswise section $x = 1.23 \text{ m}$ in Fig. 5 the results are comparable with the section at the symmetry plane, the OpenFoam

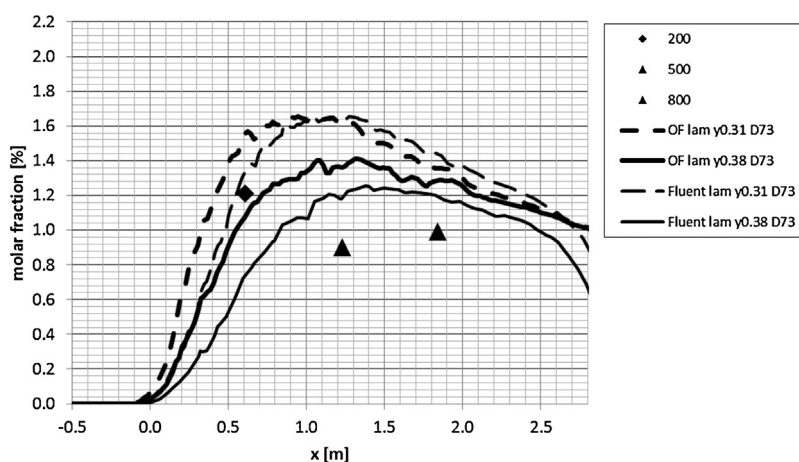


Fig. 4. Results at lateral sections for laminar calculations of DAT632.

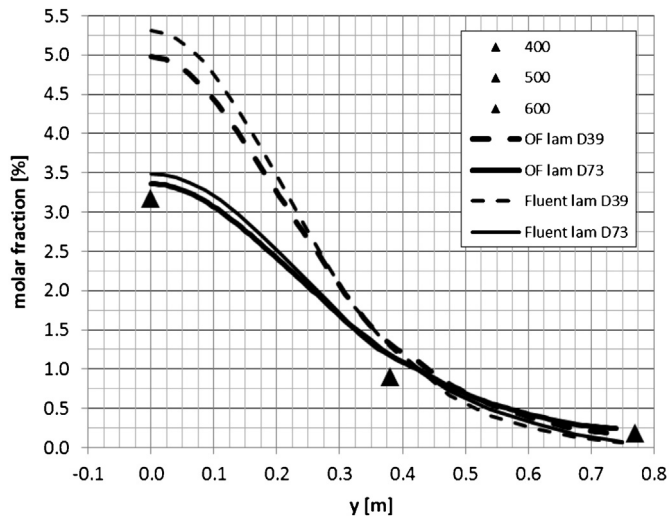


Fig. 5. Results at $x = 1.23$ m for laminar calculations of DAT632.

solution shows a slight offset which is highest at the symmetry plane; at higher y values the solutions agree very well, whereas the OpenFoam solutions show slightly more lateral dispersion of the heavy gas. The agreement with the experimental data is excellent, the absolute values and gradients are predicted correctly for the higher diffusion coefficient.

Beside the investigation of the flow topology, a statistical model evaluation was performed using simple performance measures as fractional bias (FB), geometric mean bias (MG), geometric mean variance (VG), root mean square error (RMSE) and the normalized root mean square error (NMSE). In addition, the correlation coefficient (R) and the fraction within a factor of two (FAC2) were calculated. The performance measures are defined in accordance with [25]. Another work analysing different heavy gas release studies including various numerical codes is given in [26].

For the laminar calculations, the numerical solutions with the diffusion coefficient of $D = 7.3 \cdot 10^{-6} \text{ m}^2/\text{s}$ were statistically analyzed with respect to the experimental data of the 8 available sensor data (compare Table 1). The results of the MG and VG statistics are plotted in Fig. 6. Results that show no random scatter but under or over predict the experimental data would be placed on the parabola. This parabola presents the lowest possible value of VG for a corresponding MG. Values of MG lower than 1 indicate an over prediction, larger than 1 an under prediction of the experimental results. The OpenFoam solutions show a slight over

Table 1

Statistical performance measure values for DAT632 (laminar and turbulent solutions).

	FB	MG	VG	RMSE	NMSE	R	FAC2
OF laminar	-0.155	0.829	1.051	0.46	7.93	0.995	8/8
F laminar	-0.152	0.981	1.141	0.57	9.77	0.996	7/8
OF turbulent	-0.189	0.823	1.054	0.60	10.34	0.995	8/8
F turb. D39	0.174	1.22	2.16	1.09	18.82	0.806	3/8

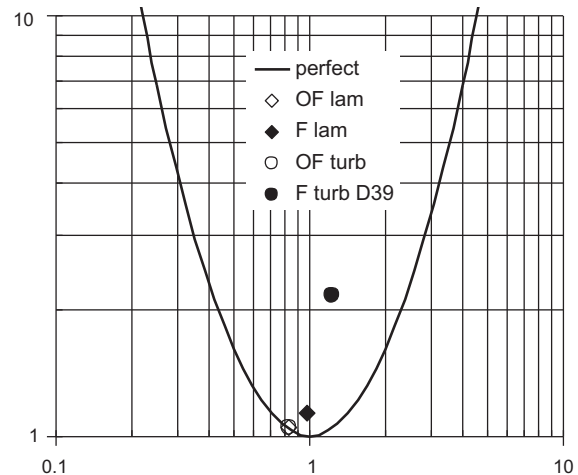


Fig. 6. Statistical analysis results of geometric mean bias (MG) and geometric variance (VG) for steady state concentration predictions of DAT632 (numerical results of OpenFoam and Fluent).

prediction of the experimental data but a very low, almost perfect variance, whereas the Fluent solution predict the geometric mean slightly better but has a higher VG. For the Fluent solutions, one data point of the numerical results is outside the range of a factor of two. Values for RMSE and NMSE show slightly better values for OpenFoam and the correlation coefficient is very close to 1 for all results. Therefore, the agreement between the numerical results and the experimental data can be considered as good for laminar, steady state values. Since the present investigations were performed without applying a turbulence model, this will be investigated in the following section.

3.2.2. DAT 632: Turbulent solutions

As shown before, the flow condition is quasi laminar, therefore it is a good test case to validate the performance of the turbulence modelling in the transition regime between laminar and turbulent

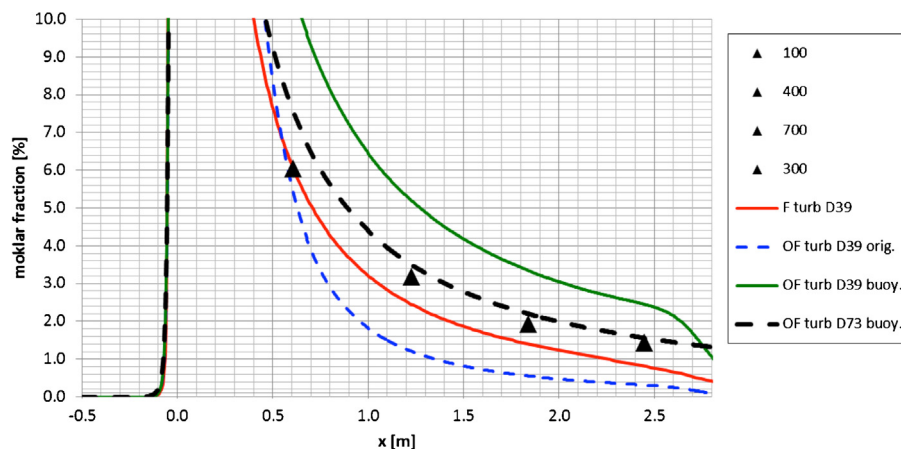


Fig. 7. Results at symmetry plane for turbulent calculations of DAT632.

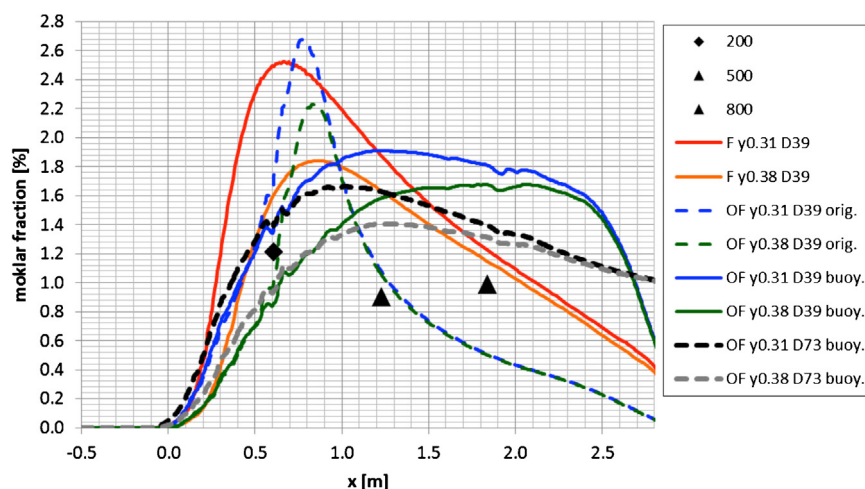


Fig. 8. Results at lateral sections for turbulent calculations of DAT632.

flow. The turbulence model should be able to predict this state by limiting the production of the turbulent viscosity. With the present heavy gas release, the turbulence created in the shear layer is expected to be damped by the vertical negative density gradient in the gas mixture. Typical heavy gas releases take place in a turbulent atmospheric boundary layer, therefore the standard $k-\varepsilon$ model was chosen since this is one of the most commonly used models in this kind of environment and also has to be able to predict the turbulence damping correctly.

Taking the Fluent solution as reference, it can be seen in Figs. 7–9 that the turbulence is over predicted which results in a faster mixing in the symmetry plane (under prediction of concentrations) and too strong lateral mixing of the heavy gas (over prediction of concentrations at higher y -values). The flow topology and concentration gradients hardly change increasing the mesh resolution which was investigated by a grid refinement study. Since the lateral spreading is even stronger with the higher diffusion coefficient, the comparison for Fluent was only performed with the lower laminar diffusivity ($D = 3.9 \cdot 10^{-6} \text{ m}^2/\text{s}$).

The OpenFoam solution with the original $k-\varepsilon$ implementation ("OF orig." in the figures) also over predicts the turbulent viscosity resulting in an over predicted turbulent mixing in both stream wise and lateral direction. Particularly, the distributions at $y = 0.31 \text{ m}$ and $y = 0.38 \text{ m}$ do not predict the gradients correctly.

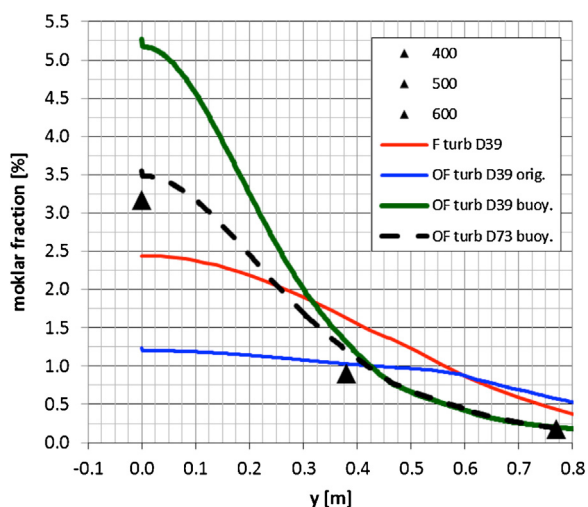


Fig. 9. Results at $x = 1.23 \text{ m}$ for turbulent calculations of DAT632.

Applying the $k-\varepsilon$ turbulence model including the buoyancy term ("OF buoy") reduces the turbulence to almost laminar level. The solution is very close to the laminar solution, compare the previous chapter. This is in agreement with the RNG solution of Fluent which is almost identical with the laminar ones (not shown here). Nevertheless it is not explainable why the Fluent $k-\varepsilon$ model strongly over predicts the turbulent viscosity since the implementation of the buoyancy term in OpenFoam in the present paper is exactly the same as described in the Fluent documentation for the standard $k-\varepsilon$ and RNG turbulence model. Obviously there is a different implementation in Fluent between the standard $k-\varepsilon$ and RNG model or the RNG model dampens the turbulence not due to buoyancy but due to the low-Re correction. Since the source code of Fluent is not available, this issue was not further investigated.

A second calculation with the higher diffusion coefficient of $7.3 \cdot 10^{-6} \text{ m}^2/\text{s}$ was performed with OpenFoam which is also shows an almost complete damping of the created turbulence due to the heavy gas. The results agree well with the ones from the laminar calculations and the experimental data.

The results of the statistical analysis of the turbulent calculations are also included in Table 1 and Fig. 6. The turbulent solution (diffusion coefficient $7.3 \cdot 10^{-6} \text{ m}^2/\text{s}$) calculated with OpenFoam show in principle the same performance as for the laminar solutions. For Fluent, only the solution with a diffusion coefficient of $3.9 \cdot 10^{-6} \text{ m}^2/\text{s}$ is shown since the higher diffusivities would result in even poorer performance. The geometric mean bias is still quite good, showing a slight under prediction of the experimental data. The variance has increased to values larger than two, being also a result that only 3 out of 8 points lay within a factor of 2 of the experimental data. As explained earlier the dispersion is over predicted resulting in an inaccurate prediction of the flow topology.

It shall be mentioned that for the calculations with OpenFoam, there is hardly any influence of applying the buoyancy term only in the k -equation (comparable to Fluent) or in the k - and also ε -equation (original version of the $k-\varepsilon$ model, not shown here). The concentrations at the symmetry plane are slightly increased by approximately 0.2–0.4%, reducing the mixing in lateral direction. Since these differences are small, they are considered as minor.

Considering this test case in a low turbulence environment, the results of OpenFoam with the implementation of the buoyancy term in the standard $k-\varepsilon$ turbulence equations in OpenFoam can be considered as reliable also for a low turbulence environment since it predicts the damping of the turbulence due to the density gradient in a physical way.

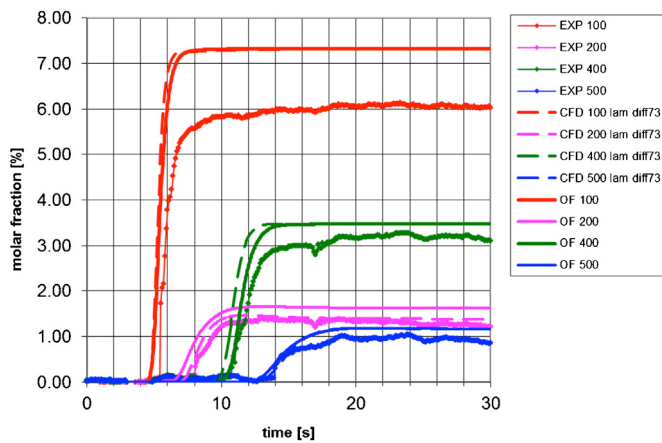


Fig. 10. Time dependent concentration at sensor positions (sensor 100, 200, 400, 500).

3.2.3. DAT 632: Time accurate solutions

Finally, the accuracy of the solvers with respect to the temporal behaviour is checked. For the Fluent solutions, a 2nd order implicit dual time stepping scheme is applied with a physical time step size of 0.05 s whereas with OpenFoam a 1st order Euler global time-stepping is applied ensuring a maximum local CFL number of 1. The timesteps are in the order of 0.002 s; calculations were performed in laminar flow without applying a turbulence modelling.

The time accurate results (diffusion coefficient $7.3 \times 10^{-6} \text{ m}^2/\text{s}$) of the sensor locations are compared with the experimental results in Figs. 10 and 11. A good agreement between the Fluent and OpenFoam calculations can be stated. The cloud arrival times, the maximum gradient and the steady state values are well predicted by the different codes. The differences in the steady state values were discussed in the previous chapter. Minor differences are present at sensor position 700 and 300, where the arrival times are identical but the Fluent solution shows a slightly steeper gradient.

The turbulent solutions of OpenFoam including the buoyancy term in the turbulence model are almost identical with the laminar solution, therefore the results will not be discussed separately. With higher turbulent viscosities (e.g. the Fluent $k-\epsilon$ model or the original $k-\epsilon$ implementation in OpenFoam neglecting the buoyancy term) the cloud arrival times are strongly overpredicted, therefore they are not further discussed here.

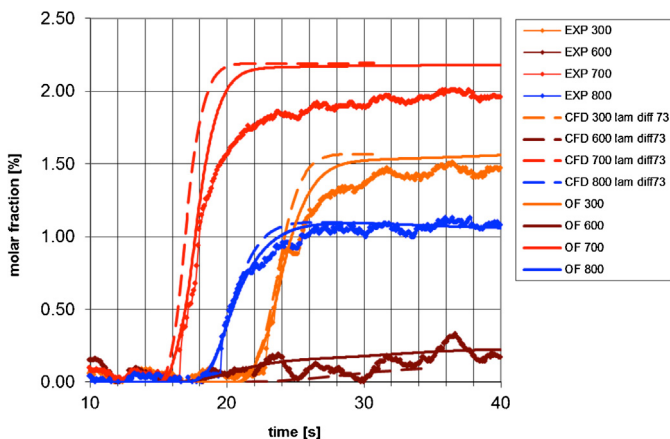


Fig. 11. Time dependent concentration at sensor positions (sensor 300, 600, 700, 800).

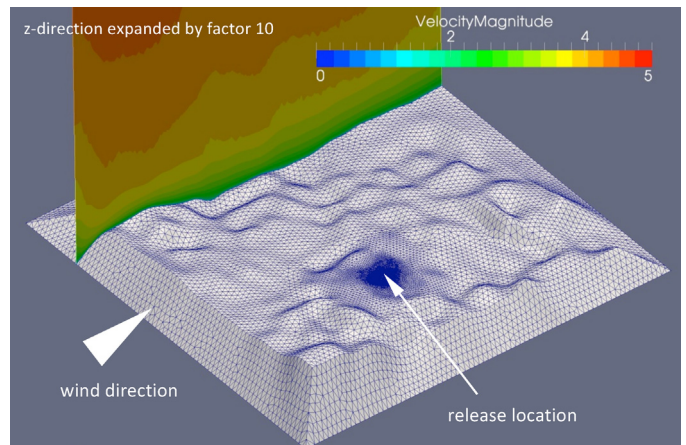


Fig. 12. Computational domain with surface mesh and vertical plane showing velocity contours (vertical direction expanded by factor 10).

4. Validation ReactingFoam for dense gas release in atmospheric boundary layer including terrain effects

4.1. Case description continuous heavy gas release over hilly, sloping terrain

In the present chapter, a code to code comparison will be performed in a full scale high turbulent environment, which consists of a hypothetical heavy gas release in an atmospheric boundary layer. Here, the full range of turbulence levels is present: almost laminar flow in the heavy gas close to the release and a strongly turbulent flow in the atmospheric boundary layer. Therefore, especially the damping of the turbulence due to the presence of the heavy gas is the most important issue. In addition, the surface is not flat but hilly and sloping, compare Fig. 12. The highest point is about 80 m, the lowest 30 m. The hills are in the range of 5–20 m. By this, also the terrain effects like depressions which are filled by the heavy gas influence the dispersion.

A representative CO_2 pipeline failure scenario was chosen [10], the continuous mass flow rate is 8821 kg/s at a temperature of 200 K. In first instance, an infinite release was chosen which reduces the computational costs significantly since a steady state conditions can be applied.

In order to model the atmospheric boundary layer, the inflow boundary conditions were chosen applying consistent Dirichlet boundary layer profiles for velocity, k and ϵ :

$$u(z) = \frac{u^*}{\kappa} \ln \left(\frac{z + z_0}{z_0} \right) \quad (9)$$

$$k(z) = \frac{u^{*2}}{\sqrt{C_\mu}} = \text{const.} \quad (10)$$

$$\epsilon(z) = \frac{u^{*3}}{\kappa(z + z_0)} \quad (11)$$

The ground boundary conditions were modelled as rough turbulent walls, relating the aerodynamic surface roughness length z_0 and the equivalent sand grain roughness of the atmospheric boundary layer k_s as:

$$C_s = \frac{E \cdot z_0}{k_s} \quad (12)$$

with E being the modelling constant for smooth walls ($E = 9.8$). The roughness constant C_s is chosen at $C_s = 9.7$ to minimize the distance of the near wall point (y_0) to be slightly larger than the aerodynamic roughness length z_0 which is a requirement of the wall function.

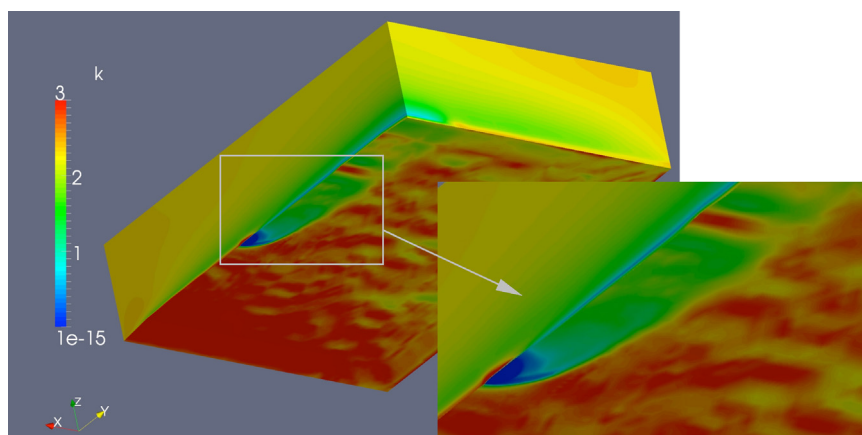


Fig. 13. Turbulence kinetic energy of Fluent reference solution.

This approach is commonly used in atmospheric boundary layer modelling, described in [27].

The atmospheric boundary layer at the inflow boundary is characterized by a velocity of 12 m/s at 10 m height following a log-law (constant velocity boundary condition). The surface roughness length z_0 is 1.74 cm (chosen according to the terrain vegetation which is grass and bush covered) and the friction velocity u^* is 0.8 m/s. The ambient air temperature is 300 K.

The computational domain is 2400 m $\times$ 2400 m $\times$ 600 m, the heavy gas is released from a 60 m circular hole as a low momentum, homogenous mass flow inlet of 8821 kg/s. At the release, a mass flow boundary condition is set and the values for k and ε are fixed at 10^{-6} m²/s² and 10^{-4} m²/s³. At the outlet, with Fluent pressure-outlet and with OpenFoam zero-gradient boundary conditions were chosen. The lateral sides of the domain are modelled as symmetry and the top boundary as a constant velocity boundary.

The topology information is taken from GIS point data (consisting of x , y , z coordinates) at a resolution of 50 m. The point data is read into CAD and a polynomial 2D-surface is created through the points. In all lateral directions, a 200 m ramp was created to reach a uniform ground level for all lateral boundaries. From this surface, the computational domain is constructed by vertical extrusion. The grid contains 600,000 cells, the first spacing is about 0.04 m and the grid size on the surface is varying from 0.5 m close to the release to 30 m at the farfield, compare also Fig. 12. The maximum cell aspect ratio is 450, the maximum grid skewness is 1.3 and the mesh non-orthogonality is 13.8 (averaged) with a maximum of 80.

The ramps between the hilly surface and the lateral boundaries do not form an issue for the code to code comparison between Fluent and OpenFoam since both codes perform the calculations on identical grids. Due to this, and of course the terrain itself, the atmospheric boundary layer will not be transported smoothly through the domain but will adapt itself to the terrain as is would do in open field or when placed into a wind tunnel.

4.2. Code to code comparison steady state solution

As discussed earlier, the k - ε type turbulence models implemented in OpenFoam don't feature buoyancy effects. This implies that without adding the buoyancy term in the turbulence model OpenFoam will fail to predict this type of flow. Initial tests highlighted the problem: The turbulence was hardly influenced by the presence of the heavy gas such that the mixing of the heavy gas was strongly over predicted and the concentrations decreased too rapidly. Therefore, only the solutions including the buoyancy terms will be discussed.

The treatment of the buoyancy-term in the ε -equation is still a point of discussion in heavy gas applications. In order to match specific flow features, most of the commercial codes have individual treatment of the buoyancy term in the ε -equations [19,20]. Since Fluent recommends to exclude the buoyancy term in the ε -equation, the Fluent reference solution were performed with the corresponding setting. As mentioned before, all computations were performed on identical meshes.

The following Figs. 13 and 14 are all plotted as half domains with the symmetry plane in the release; the view is from below on the ground and symmetry plane of the domain.

The distribution of the turbulence kinetic energy of the reference solution of Fluent is plotted in Fig. 13; the damping of the turbulence can be seen clearly downstream the released heavy gas. Typical heavy gas horse shoe vortices can be observed originating from the leading edge of the interface between the heavy gas at the release location and the atmospheric boundary layer. The prediction of these thin vortical structures of low turbulence is crucial for an accurate dispersion prediction, since this mainly transports the heavy gas in lateral direction. In addition, turbulence is created at the shear layer between the heavy gas and the atmospheric boundary layer which increases the mixing.

It shall be mentioned that the minimum values of the turbulence kinetic energy of the heavy gas region are not achieved directly on the ground but in some distance (approximately at half the distance between the upper shear layer to the ground) since the ground surface re-creates turbulence in the wake of the mixing heavy gas.

Away from the heavy gas, the surface topology leads to a pattern of variable turbulence values on the ground according to the hilly surface geometry.

The results of OpenFoam are given in Fig. 14. The first solution without the buoyancy term in the ε -equation is in good agreement with the Fluent reference solution, all features discussed above are reproduced qualitatively and quantitatively. With Fluent, only a minor difference was seen between the solutions excluding or including the buoyancy term in the ε -equation. With OpenFoam including buoyancy term in the ε -equation (according to the original formulation) shows a much larger effect of the dispersed heavy gas (Fig. 14). Heavy gas is spread more in lateral direction and the mixing is reduced due to the increased dissipation of turbulence inside the heavy gas. These effects will be discussed later.

The surface distributions of k , ε , temperature and molar fractions of CO₂ are given in Figs. 15–18 for the Fluent (left figures) and OpenFoam solutions excluding the buoyancy term in the ε -equation (middle figures). A very good agreement is given for all the values including turbulence kinetic energy, turbulence dissipation rate, CO₂ molar fractions and temperature. The dissipation distributions

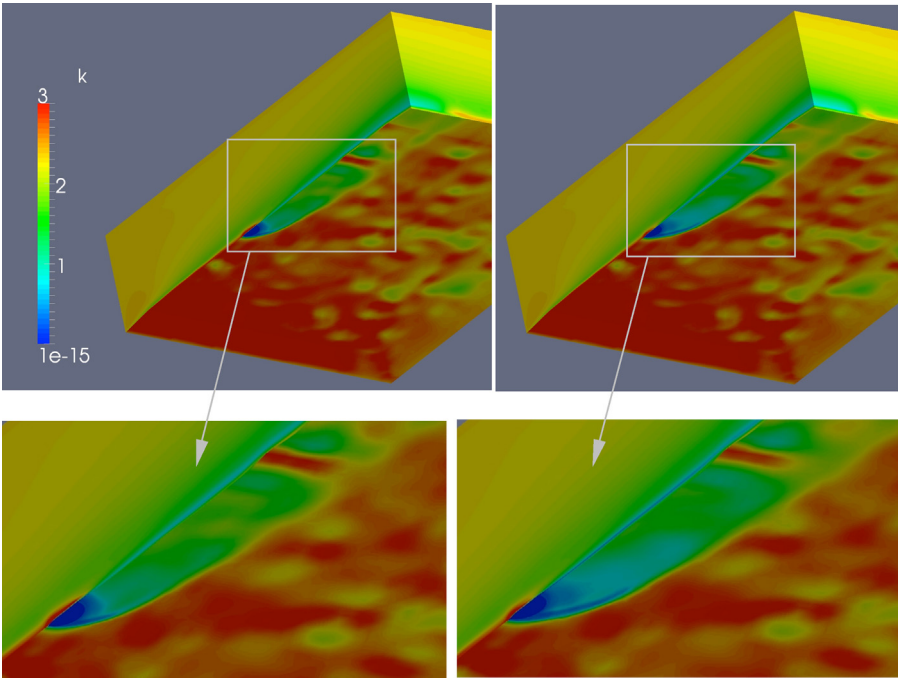


Fig. 14. Turbulence kinetic energy of OpenFoam solution: buoyancy term excluded (left) and included (right) in dissipation equation.

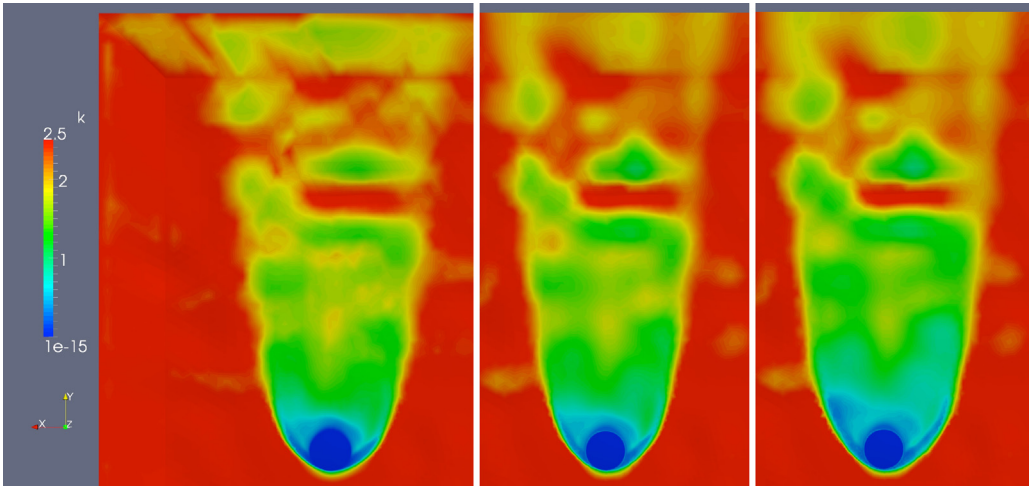


Fig. 15. Surface turbulence kinetic energy: Fluent (left) OpenFoam (middle) OpenFoam including buoyancy term in ϵ -equation (right).

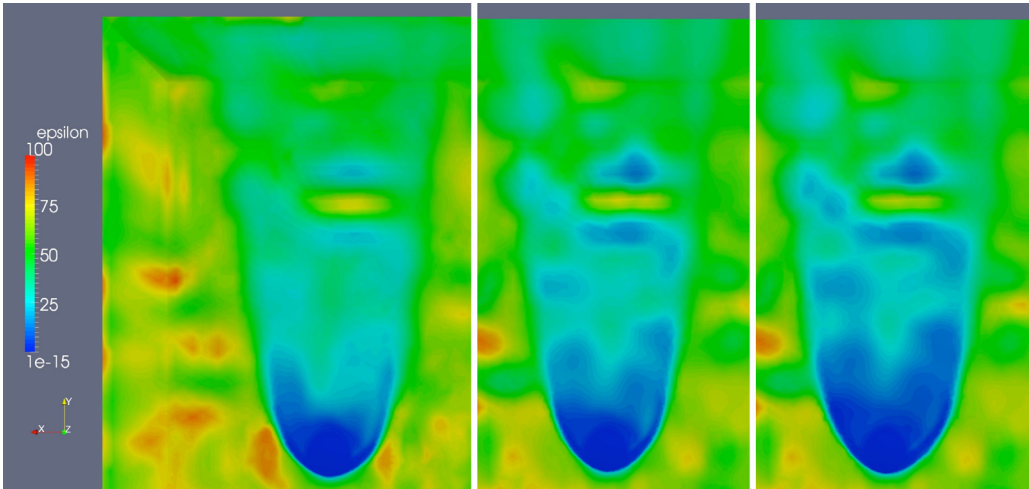


Fig. 16. Turbulence dissipation: Fluent (left) OpenFoam (middle) OpenFoam including buoyancy term in ϵ -equation (right).

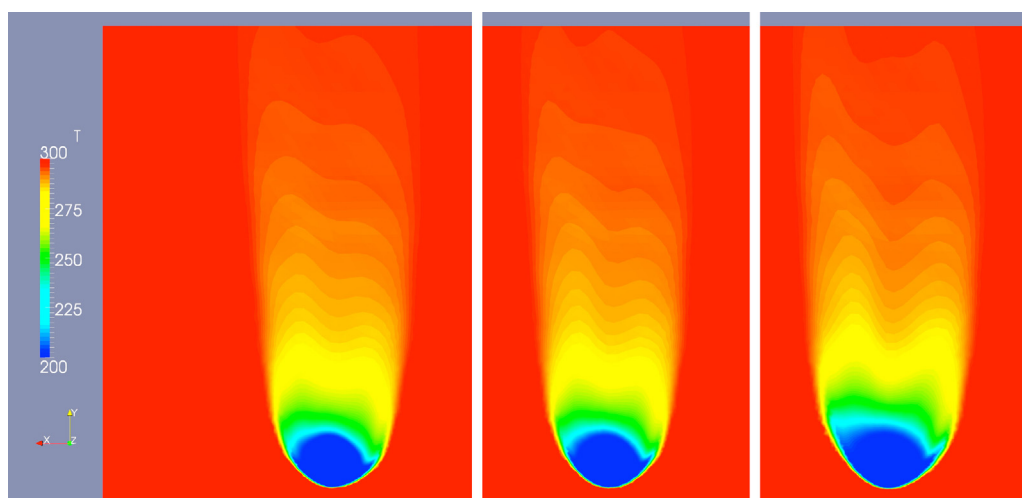


Fig. 17. Surface temperature: Fluent (left) OpenFoam (middle) OpenFoam including buoyancy term in ε -equation (right).

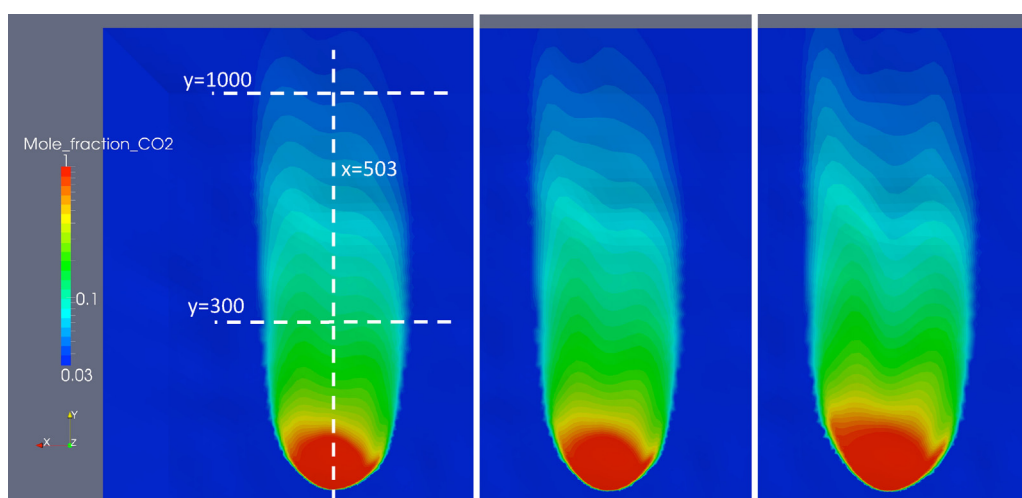


Fig. 18. Surface molar CO_2 fractions: Fluent (left) OpenFoam (middle) OpenFoam including buoyancy term in ε -equation (right).

are plotted in the first cell from the ground. Clear effects of the surface topology can be seen especially in the k and ε values (varying wall shear stresses), whereas in the temperature and molar fractions only minor differences are present by slightly asymmetric distributions. Also these minor features are reproduced by both codes consistently. Especially the flow topologies close to the release area are in very good agreement, showing the typical heavy gas shape of the released gas and a significant damping in turbulence. The results of OpenFoam including the buoyancy term in the ε -equation show stronger heavy gas behaviour, the lateral dispersion is increased and the mixing reduced like mentioned before. Especially with the distributions of temperature and molar fractions, the heavy gas behaviour is present also at larger distances from the release.

Finally, some sections in flow and lateral direction are discussed for the most interesting values for external safety issues, distributions of molar fraction and temperature at sections downstream the release ($x=503$ m, Figs. 19 and 20) and at lateral sections at 300 m (Figs. 21 and 22) and 1000 m (Figs. 23 and 24) downstream the release. The locations of the sections are also plotted in the previous Fig. 18. Solutions are given from Fluent ("Fluent"), OpenFoam with the original settings ("OF orig.", turbulent Prandtl and Schmidt number of 1.0, 1st order upwind discretizations scheme) and OpenFoam with modified parameters identical to the Fluent settings ("OF mod.", turbulent Prandtl number 0.85 and turbulent

Schmidt number of 0.7, 2nd order upwind discretizations scheme). The solution of OpenFoam with modified settings agrees well with the Fluent reference solution at $x=503$ m. The importance of using identical discretization schemes and settings can be seen when comparing these solutions with the OpenFoam solution with original settings which was performed at different turbulent Schmidt and Prandtl numbers and lower order scheme where the mixing of the heavy gas is clearly reduced. The emphasis here is

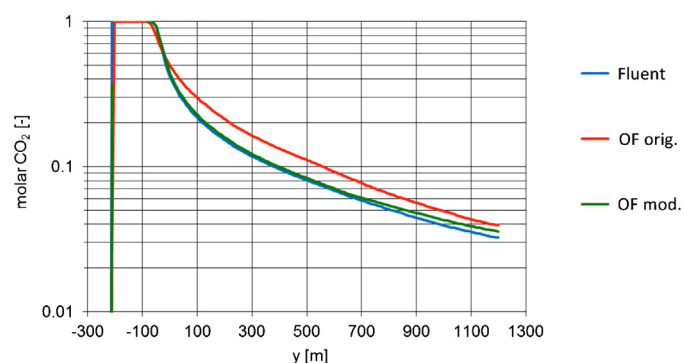
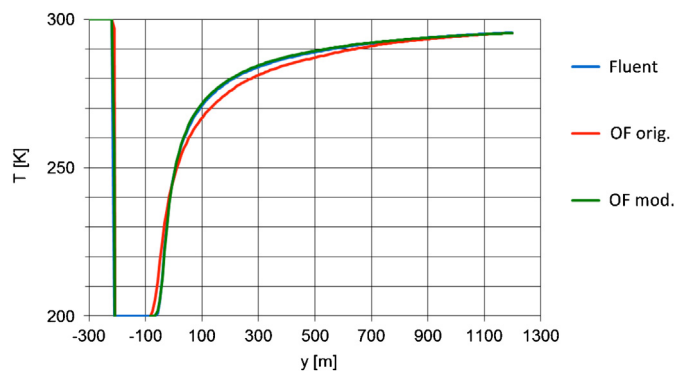
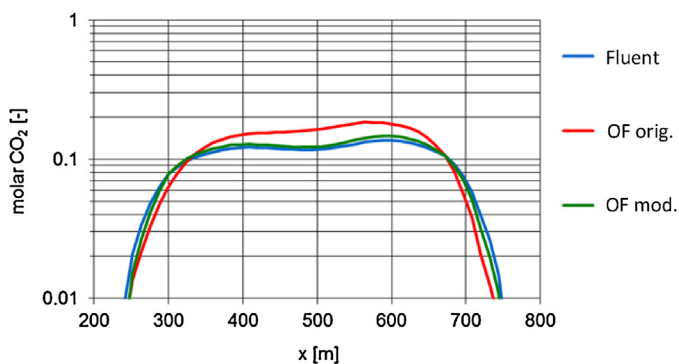


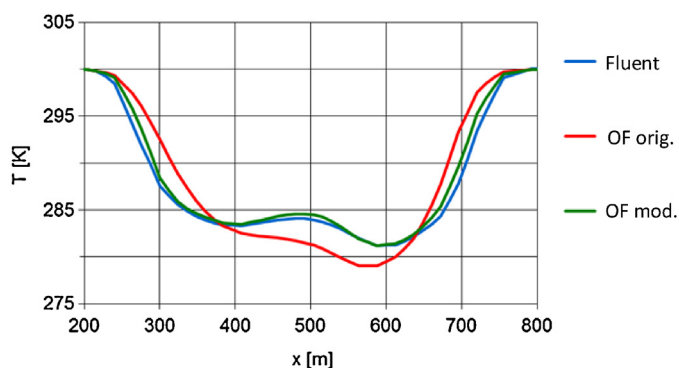
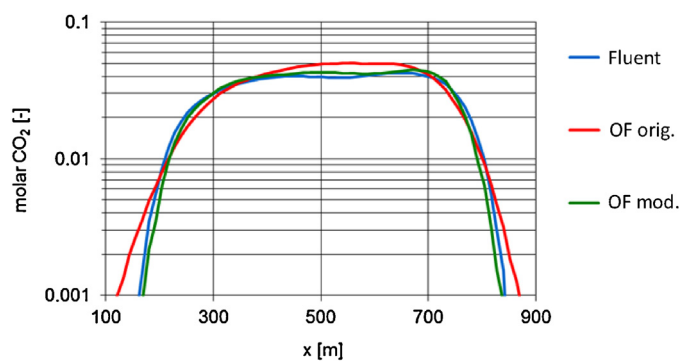
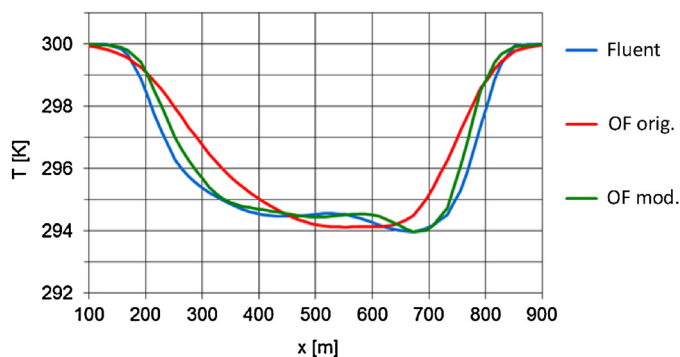
Fig. 19. CO_2 molar fraction distribution at $x=503$ m.

Fig. 20. Temperature distribution at $x = 503$ m.Fig. 21. CO_2 molar fraction distribution at $y = 300$ m.

on the results of calculations performed with OpenFoam following best-practice from tutorials at robust settings (first order scheme) without the ability of adjusting turbulent momentum and heat transfer diffusivity rather than on analyzing the effects of the discretization scheme and turbulent diffusivities separately.

At $y = 300$ m (Figs. 21 and 22) and 1000 m (Figs. 23 and 24) the results are of the same quality. Applying the modified settings, there is an excellent agreement between Fluent and OpenFoam, all details of the flow are represented comparably. As can be seen clearly, the solution with original settings is of lower accuracy and has a reduced mixing due to different settings explained before.

Overall, comparing with Fluent, OpenFoam predicts slightly reduced mixing resulting in slightly higher molar fractions at the plane downstream the release and slightly narrower dispersion patterns. The differences at the plume edges in molar fractions are about 0.2%, the difference in temperature about 1 K at $y = 1000$ m from the release. Since the mesh resolution here is roughly 30 m, this is also correlated to the discretization and the numerical

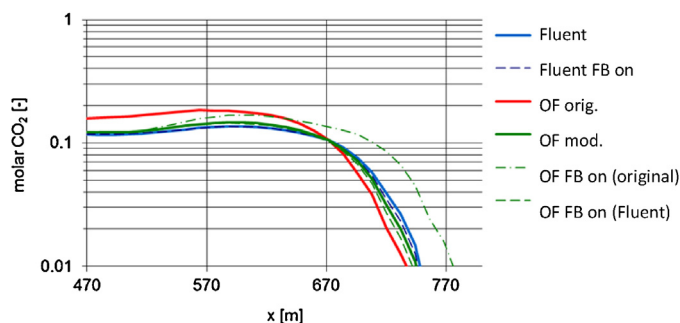
Fig. 22. Temperature distribution at $y = 300$ m.Fig. 23. CO_2 molar fraction distribution at $y = 1000$ m.Fig. 24. Temperature distribution at $y = 1000$ m.

diffusion of the schemes. Once again, these differences are considered as minor and the codes show an excellent agreement for this test case.

As mentioned earlier, there are different ways applying the buoyancy term in the turbulence model. A comparison between the results of the different treatments is shown in Fig. 25. Here, only a detail is given at $y = 300$ m, showing only the right part of the domain for clarity.

Applying the buoyancy term in the ε -equation is called "FB on" (= "full buoyancy on", the Fluent notation), the implementations in OpenFoam are called "original" (from the standard k - ε -model, Eq. (6)) and "Fluent" (the Fluent implementation, compare Eq. (8)).

As explained earlier, the OpenFoam ("OF mod.") and Fluent solutions ("Fluent") are in good agreement. Applying the buoyancy term in the ε -equation according to the Fluent formulation results in a very small difference for both solutions (Difference between "Fluent" and "Fluent FB on" and "OF mod." and "OF FB on (Fluent)"). Both Fluent and OpenFoam result in slightly reduced lateral dispersion,

Fig. 25. Effect of buoyancy term in diffusion equation on CO_2 dispersion distribution ($y = 300$ m).

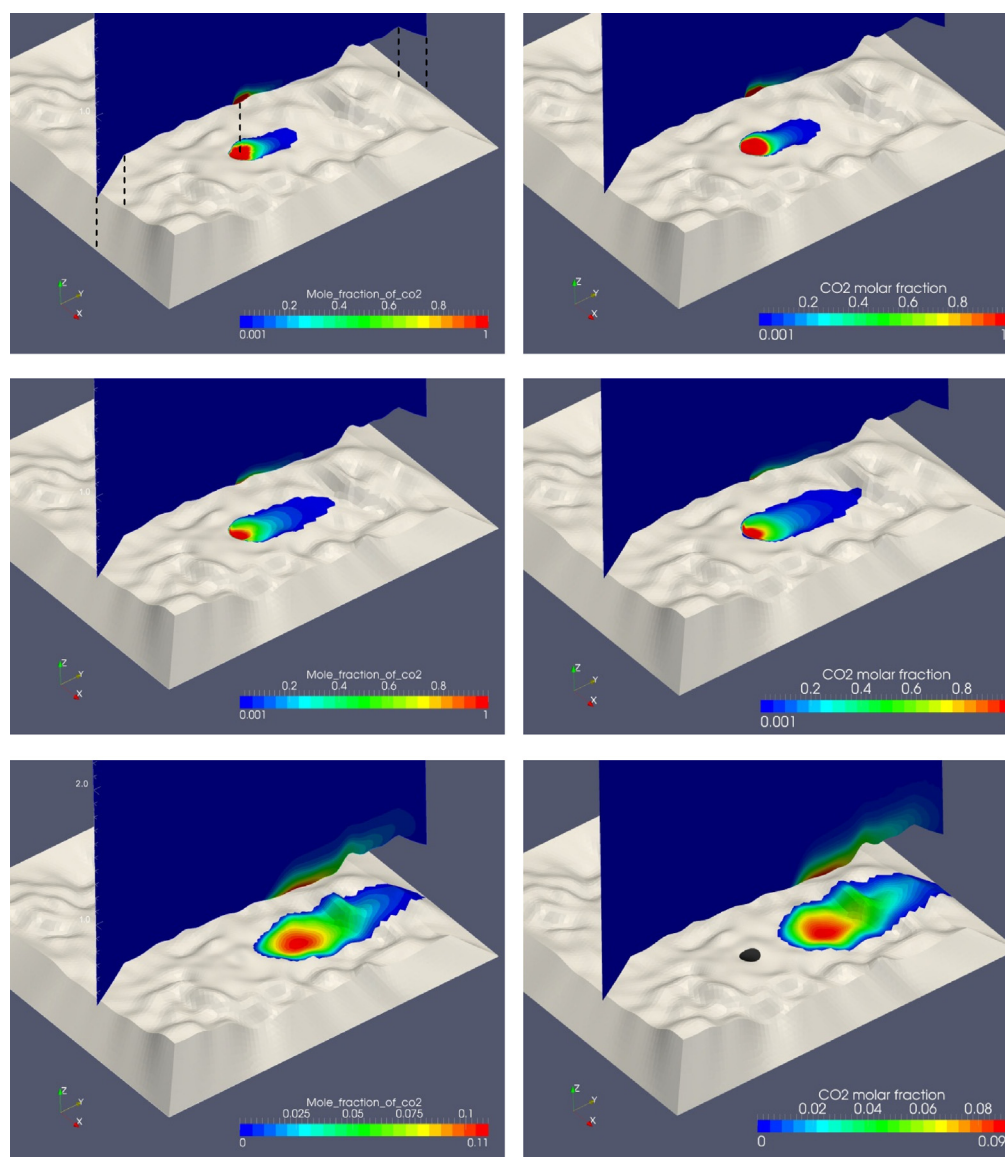


Fig. 26. Comparison of unsteady dispersion results of Fluent (left) and OpenFoam (right) at time steps of 34 s, 55 s and 105 s (top to bottom); molar fractions on ground and vertical plane through release point (vertical direction scaled by factor 10; plane shifted vertically for better visibility).

by this it can be demonstrated clearly that the implementations result in a comparable behaviour.

A major difference is observed only in comparison with the model implemented according to the original $k-\varepsilon$ -model ("OF FB on (original)") where the lateral dispersion is much higher and the overall mixing process delayed. Here, the plume is roughly 10% wider since the heavy gas behaviour is stronger due to increased turbulence destruction. Since the present investigation is just a numerical code to code comparison without experimental data, the correct modelling of the dissipation for heavy gases should be investigated further in the future.

4.3. Code to code comparison time accurate solution

In the high turbulence environment, the comparison for time accurate calculations shall be performed since with the present case, quite strong sensitivities are expected. Applying a semi-continuous release with a mass flow rate of 8821 kg/s over 34 s [10], the dispersion takes place over the hilly terrain where the flooding and venting of depressions is more sensitive in combination with

the spilled finite mass. The grid and flow conditions are identical with the previous study for steady state conditions.

With Fluent, an implicit 2nd order dual time stepping scheme, with OpenFoam a 1st order Euler global time stepping scheme was applied. The physical time steps were 1 s in Fluent and 0.005 s in OpenFoam which is limited due to the global time stepping to the local CFL number. Since the timestep in OpenFoam is two orders of magnitude lower than with Fluent, the accuracy of the first order time stepping scheme applied here is considered sufficient.

The results at the discrete times of 34 s (end of the spill), 55 s and 105 s are shown in Fig. 26. In addition to the surface, the molar fractions are plotted as well in a vertical streamwise plane through the release point such that the vertical entrainment of the dispersed gas can be compared. The solutions at the different time steps agree qualitatively well which means the shape and the position of the plume are in good agreement. The OpenFoam solution seems to advance in time slightly faster, this can be seen at the lowest molar fractions at the end of the plume which are slightly further advanced in location at the same timestep (55 s). Also the maximum molar fractions are slightly higher with Fluent (11% and

9% OpenFoam at $t = 105$ s) and the heavy gas entrainment into the atmosphere seems to be slightly less. This might be the influence of the physical timestep in Fluent which was chosen at 1 s due to the reduced computational effort. Already at 34 s, the end of the release of the heavy gas, there are minor differences visible which might be an indication that the initial release process is slightly different. Since the initial sudden release is strongly unsteady, the chosen timestep might not sufficiently resolve these processes. With too high timesteps, the overall dispersion processes are usually delayed and also initial differences propagate through time. In addition, turbulence modelling, discretizations errors, etc. might have a contribution. Nevertheless the agreement is good for both codes also for time accurate simulations in a high turbulence environment.

5. Summary

In the present paper, the OpenFoam software was extended also to be able to predict heavy gas dispersion in atmospheric boundary layer flows. In a low turbulence, gravity driven environment, OpenFoam was validated against experimental data and it could be demonstrated that the damping of the turbulence due to the presence of the heavy gas is modelled correctly by the turbulence model. The standard $k-\epsilon$ turbulence model was applied and the agreement with the experimental data was excellent. A code to code comparison was performed with the commercial Fluent software, an excellent agreement was found for the laminar calculations for steady and time accurate results, but the $k-\epsilon$ model implemented in Fluent failed to dampen the turbulence due to the density gradient. State of the art validation metrics were applied to assess the quality of the CFD solutions. OpenFoam shows a slight over prediction at almost perfect variance for both laminar and turbulent calculations. The Fluent solution have better mean values but at higher variation (laminar solutions). For the turbulent solutions of Fluent, both mean and variation are poor. Since the flow topology prediction is a main issue, the OpenFoam solution with almost perfect variance is considered better since it results in a much better prediction of the gradients in laminar and turbulent flow.

A second numerical test case for code to code comparison was chosen in an atmospheric boundary layer flow. In a full-scale sloping, hilly terrain CO₂ was released. The interaction of the terrain, gravitational and heavy gas effects on the turbulence model were found to be in excellent agreement between both codes. For a time accurate case, the Fluent solutions appeared to be dispersed slightly slower.

Based on the present work it can be summarized that the modified version of OpenFoam is able to predict heavy gas dispersion in a wide range of turbulence in at least the same quality compared with commercial codes.

Acknowledgements

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